

Model Immunization from a Condition Number Perspective

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Title Illustration Placeholder

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Intro Slide 1

Motivation

Intro Slide 2

Abstract & Motivation

- Goal: pre-train models that resist fine-tuning on harmful tasks while preserving utility on benign tasks.
- Key idea: analyze and control **Condition Number** of the fine-tuning Hessian.
- Provide a framework for linear models and an algorithm with condition-number regularizers.
- Empirically effective on linear regression/classification and deep nets (ResNet, ViT).

Condition number closer to 1 → easier optimization; larger → slower convergence for gradient-based methods.

Introduction

- Immunization (Zheng & Yeh, 2024): hinder malicious fine-tuning; mitigate misuse of open-source models.
- Prior work: empirical on text-to-image; lacked precise definition and conditions for success.
- Our perspective: **condition number** of Hessian during linear probing determines fine-tuning difficulty.
- Geometric insight: angles between singular vectors of pre-training vs. harmful covariances matter; metric **RIR**.

Contributions

- Framework using **condition numbers** to define and evaluate immunized models.
- Two regularizers: decrease (**R_well**) and increase (**R_ill**) condition numbers with monotonic guarantees.
- Algorithm integrating task loss + dual regularizers; validated on linear and deep models.

Metric

RIR >> 1

Relative Immunization Ratio

Tasks

Linear + Deep

Regression · Classification · ResNet · ViT

Preliminaries: Condition Number & GD

- **Condition number** $\kappa(S) = \sigma_{\max}(S) / \sigma_{\min}(S)$ (pseudoinverse when needed).
- For strongly convex L with Hessian H : GD convergence slows as $\kappa(H)$ increases.
- Regularization idea: shape $\kappa(H)$ of fine-tuning problems to control optimization difficulty.

Formula placeholders for $\kappa(S)$ and GD rate

Transfer Learning via Linear Probing

- Freeze feature extractor f_θ ; learn linear classifier h_w over target data D .
- Optimization: minimize task loss (e.g., ℓ_2 , **cross-entropy**) over w .
- Focus: harmful task D_H vs. pre-training task D_P .

Linear probing objective placeholder

Definition: Immunized Model (Linear Setting)

- Harder harmful probing: $\kappa(H_H(\theta_I)) \geq \kappa(H_H(I))$.
- Not harder on pre-training: $\kappa(H_P(\theta_I)) \leq \kappa(H_P(I))$.
- Maintain **D_P** performance; for invertible θ_I , equality achievable.

$H(\theta) = \theta^T K \theta$; $K = X^T X$ covariance of data.

Analysis: Hessian Spectrum & Alignment

- For linear probing with ℓ_2 loss: Hessian $H_H = \theta^T K_H \theta$.
- Singular values depend on alignment of θ and K_H singular vectors (Proposition 3.2).
- Trade-off: if K_P and K_H singular vectors align, simultaneous improvement is limited.

Diagram placeholder: angles between singular vectors (K_P vs K_H)

Objective and Algorithm (Alg. 1)

- Minimize: task loss $L + \lambda_P \mathbf{R_well}(H_P(\theta)) + \lambda_H \mathbf{R_ill}(H_H(\theta))$.
- Update ω by $\nabla_\omega L$; update θ by gradients of L , $\mathbf{R_well}$, $\mathbf{R_ill}$.
- Design ensures monotonic κ control under conditions.

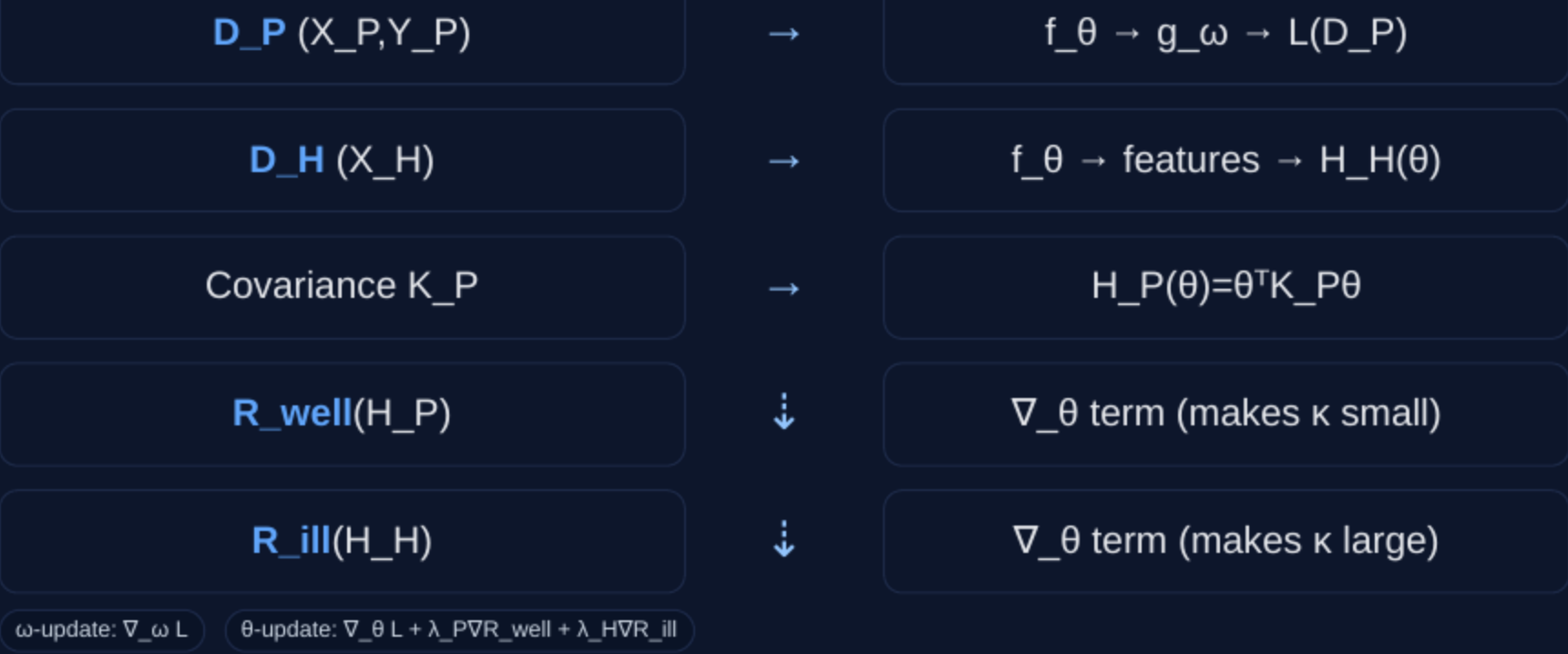
Pseudocode Sketch

```
for t = 0..T-1:  
   $\omega \leftarrow \omega - \eta \nabla_\omega L(D_P; \omega, \theta)$   
   $\theta \leftarrow \theta - \eta (\nabla_\theta L + \lambda_P \nabla_\theta \mathbf{R\_well}(H_P)$   
     $+ \lambda_H \nabla_\theta \mathbf{R\_ill}(H_H))$ 
```

Implementation Tip

Single backward pass via a dummy layer that left-multiplies grads by inverse covariances.

Flow Diagram of Training Loop



Condition-Number Regularizers

$\mathbf{R_well}(S)$: minimize $\kappa(S)$

- Differentiable; upper bounds $\log \kappa(S)$; monotonic κ decrease under GD.
- Apply to $H_P(\theta)$ to keep benign task well-conditioned.

$\mathbf{R_ill}(S)$: maximize $\kappa(S)$

- Nonnegative; upper bounds $1/\log \kappa(S)$; differentiable if $\sigma_{\min}$ unique.
- Monotonic κ increase with proper step size.
- Apply to $H_H(\theta)$ to hinder harmful fine-tuning.

For $H(\theta) = \theta^T K \theta$, gradients w.r.t. θ are available; guarantees preserved (Theorems 4.2–4.3).

Theoretical Guarantees

- Monotonic κ decrease for $\mathbf{H}_P$ under $\mathbf{R}_{\text{well}}$ updates.
 - Monotonic κ increase for $\mathbf{H}_H$ under $\mathbf{R}_{\text{ill}}$ updates.
 - Conditions: uniqueness of extreme singular values; appropriate step sizes.
- Guarantee illustration placeholder (κ over iterations)

Results: Linear + Deep Nets

Main takeaway: Ours achieves $RIR \gg 1$, making harmful probing harder while keeping pre-training well-conditioned.

- **Headline numbers:**
 - House Prices: $RIR \approx 356.2$ (Opt $\kappa \approx 92.6$; R_{ill} Only ≈ 1.24 ; IMMA ≈ 2.00).
 - MNIST (avg over 90 pairs): $RIR \approx 70.0$ with low variance (Opt $\kappa \approx 69.7$ high var).
 - Deep nets (ImageNet $\rightarrow$ Cars/Country211): strong RIR_{00} ; ImageNet accuracy maintained (ResNet slight drop; ViT improves).
- Convergence behavior: Ours slows $\mathbf{D}_H$ and speeds $\mathbf{D}_P$ under exact line search (norm-ratio), consistent with κ control.

Deep Nets: Setup and Metric

- **D_P**: ImageNet; **D_H**: Stanford Cars, Country211.
- Architectures: **ResNet18** (last 2 blocks), **ViT** (final block).
- **RIR_θ0** compares κ vs initialization; $\hat{H}$ from features **f_θ**.

Equation placeholders for $RIR_{\theta 0}$ and $\hat{H}(\theta)$

Fine-tuning Behavior on D_H (Cars)

- Linear probing accuracy vs epochs: **Ours** shows slowest convergence on **D_H**.
- Consistent with higher κ on harmful task.

Figure 3 placeholder: Test accuracy vs fine-tuning epochs

Discussion and Limitations

- Guarantees rely on unique extrema and step sizes; combined gradients may interact.
- Linear-theory gap for deep nets; empirical results are positive.
- Hyperparameter balance between $\kappa(\mathbf{H}_\mathbf{P})$ and $\kappa(\mathbf{H}_\mathbf{H})$ is important.
- Efficient implementation via graph trick; single backward pass.

Related Work

- AI safety and model misuse mitigation; unlearning vs immunization.
- Condition number in optimization; preconditioning; recent κ -regularization.
- Our novelty: principled κ -increase regularizer and immunization framing.

Conclusion

- **Condition-number** view yields precise immunization definition and metrics.
- Dual regularizers steer κ for harmful vs benign tasks with guarantees.
- Effective across linear models and deep nets while preserving utility.

Code
github.com/amberyzheng/model-immunization-cond-num

Takeaway
RIR $\gg$ 1

References (Selected)

Boyd & Vandenberghe (2004); Bubeck (2015); Golub & Van Loan (1996)
Nenov et al. (2024) – κ -minimizing regularizer
Zheng & Yeh (2024, 2025) – Model immunization (IMMA baseline; multi-concept extension)
He et al. (2016) ResNet; Dosovitskiy (2021) ViT
Deng et al. (2009) ImageNet; Krause et al. (2013) Cars; Radford et al. (2021) Country211
LeCun (1998) MNIST; Montoya & DataCanary (2016) House Prices

References (Baselines & Methods)

- Zheng, A. Y., & Yeh, R. A. (2024). Model Immunization. Empirical formulation used as baseline **IMMA**.
- Nenov, P., et al. (2024). Differentiable regularizer for minimizing matrix condition number (**R_well** inspiration).
- Opt κ : direct optimization of $\kappa(\mathbf{H}_P) - \kappa(\mathbf{H}_H)$ as described in paper's baselines.

See paper Sections 4–5 and Tables 1–3 for exact configurations and numbers.